

Suggested actions

Options for rebuilding the evaluation framework ahead of the next submission — offered as suggestions to discuss and revise, not as a settled plan.

Submission #2937 · Baseline: ARR May 2026 draft · 150 templates · 5 branches

01 — THE DIAGNOSIS

The title promises verification. The mechanism takes a vote.

Five reviewers across two cycles attacked the evaluation framework from different directions — gFWV's "validated by the same type of system," 9W1B's "shared model biases," yAYU's observation that GPT-5 is *simultaneously* a Tribunal judge and an evaluated model. These may be one objection wearing three hats. The paper is called *Verifiable Process Supervision*; verification is currently a majority vote among three frontier LLMs. Reviewers appear to detect that gap even when they don't name it, which suggests patching the Tribunal would relocate the complaint rather than resolve it.

AN ASSET THAT MAY BE WORTH EXPLOITING

EngTrace's templates are **executable Python that computes the answer**. Every intermediate quantity — every `R_A`, `v_z_max`, `F_a`, `sigma_AB` — exists as a typed value inside the generator, with known units and a known governing formula. It is then f-string-serialized into prose, that structure is discarded, and three frontier APIs are paid to approximately recover it.

If templates instead emitted a **structured trace** — symbol, value, unit, formula, dependencies — step verification could become a deterministic numeric and dimensional check against known milestone quantities, with no model in the critical path. That would be verifiable process supervision in a literal sense, it is only possible because generation is symbolic, and it could turn the framework's main liability into a contribution.

This is a hypothesis, not a recommendation to act on yet. The suggestions below are structured so it gets tested against alternatives on human-labelled data before anyone refactors 150 templates.

02 — ONE DEPENDENCY WORTH RESPECTING

A gold standard likely has to come first

Choosing between evaluators empirically requires a trustworthy reference to score them against. At present that reference is the $n = 100$ study in Appendix M — which is itself what reviewers are attacking. That suggests the human-labelled set isn't one workstream among several but a precondition for the comparisons, and it is the item with the longest lead time because it depends on people.

Two changes seem worth considering. **Size:** distinguishing $p = 0.63$ from $p = 0.75$ with reasonable power needs roughly 250–300 items rather than 100. **Ceiling:** the paper never reports human–

human agreement, which is arguably what would make $p = 0.632$ interpretable. If three experts agree with each other at $p \approx 0.70$, then 0.632 sits near that ceiling and 9W1B's "only moderate for safety-critical" reading becomes much harder to sustain. Without the ceiling, 0.632 has no reference point.

03 — SUGGESTED WORKSTREAMS

Five pieces, in roughly this order

Ordering reflects dependency rather than schedule: the gold standard gates the comparisons, and the comparisons gate any decision about instrumenting templates. Annotation and engineering could run concurrently.

01 Foundations and a frozen slice

- Recover the raw generations for all 27 models ($27 \times 1,350$) and verify integrity. Re-scoring is cheap once they're in hand; re-running inference is more costly but you've indicated it's acceptable.
- **Consider freezing one pilot slice that every candidate is tested on:** perhaps 20 templates stratified across the three original branches $\times$ three difficulty tiers, spanning known-hard domains (Transport Phenomena, Fluid Mechanics) and known-easy (Mechanics of Materials). Three instances each would give 60 problems.
- **Trace sources could span two axes at once** — capability and judge-family. GPT-5, Claude Opus 4.7 and Gemini 3.1 Pro would put all three judge families in the pool; DeepSeek R1 and Llama 3.1 70B would serve as non-judge-family controls. $60 \times 5 = \mathbf{300 \text{ traces}}$, which happens to match the gold-standard size suggested below.
- A single harness — (`evaluator_id`, `trace_set`) $\rightarrow$ `scores`, deterministic seeds, cached calls, thresholds exposed as parameters — would let every candidate be compared on identical footing, and would make the threshold sensitivity analysis fall out as a byproduct.

Would produce: a frozen pilot set · a working harness · the recovered inference archive.

02 Literature review — five suggested areas

- **Process supervision & PRMs.** Lightman et al. 2023, Uesato et al. 2022, Math-Shepherd, OmegaPRM, AutoPSV; open PRMs (Skywork, Qwen2.5-Math-PRM, RLHFlow). **ProcessBench** looks like the essential read — a meta-benchmark for step-level error identification, structurally close to what a bake-off would require. Worth answering: can an off-the-shelf PRM serve as Tier 2, and how do PRMs treat valid alternative derivations?
- **LLM-judge validity and self-preference.** Zheng et al. MT-Bench (names self-enhancement bias); Panickssery et al. 2024, "**LLM Evaluators Recognize and Favor Their Own Generations**" is probably the most directly useful citation for yAYU #1; also Stureborg et al. 2024, Bavaresco et al. 2024, Thakur et al. "Judging the Judges". Worth re-reading

Verga et al. PoLL, already cited, specifically to check whether it tests *same-family* overlap — if it doesn't, that gap may be claimable.

- **Deterministic and verifiable evaluation.** GSM-Symbolic, functional benchmarks, MathCheck, the RLVR turn toward deterministic checkers, CAS-based equivalence in math grading, unit and dimensional checking, and autoformalization (Lean/Isabelle) as an extreme worth positioning against rather than adopting.
- **Meta-evaluation methodology.** Deutsch et al., "A Statistical Analysis of Summarization Evaluation Metrics" is the standard treatment of CIs and power for metric meta-evaluation, and speaks directly to yAYU #3. Also worth looking at agreement ceilings as normalization, and **Williams' / Steiger's test for dependent correlations** — arguably the correct test for " $p = 0.632$ beats $p = 0.325$ on the same sample," which two independent CIs would not establish.
- **Scoring practice in science & engineering benchmarks.** SciBench (5%), ABench-Physics (1%), UGPhysics, PhysReason, EngiBench, TransportBench, EEE-Bench — who checks process at all, and how.

Would produce: an annotated bibliography · a positioning table for §2 · confirmation or revision of the candidate set below.

03 A human gold standard

- **Two label levels may both be needed.** *Trace-level:* reasoning quality 1–5 on the existing Table 10 rubric, for correlating against evaluator scores. *Step-level:* each step marked correct / correct-but-alternative-path / incorrect, plus first-error position. The step-level labels would be the addition — trace-level alone shows that an evaluator disagrees but not *where*, and "alternative path" is precisely the category any deterministic candidate must be shown not to punish.
- **Roughly 300 traces, ≥ 2 experts each, with a ~100-trace triple-labelled overlap** for the ceiling and for IAA (Gwet's AC1 on step labels, ρ on trace scores).
- **Reporting the human–human ceiling** costs nothing extra to obtain and would reframe every downstream result.
- Blind to model identity, randomized presentation order, and a few seeded gold-anchor items to detect annotator drift.

Would produce: dual-level labelled traces · a human–human ceiling · IAA with CIs · the reference every later claim is measured against.

04 Candidate evaluators to compare

Six options, all runnable through one harness. E0 is included because without an unchanged baseline the comparisons have no anchor.

SYMBOLIC deterministic, no model in the loop **HYBRID** deterministic first, model on residuals

ID	MECHANISM	TYPE	COST	WHAT IT MIGHT BUY
E0	Current framework, unchanged — Tier 1 numeric + cross-encoder, Tier 2 three frontier judges	JUDGED	none	An honest baseline. Without it, every comparison is unanchored.
E1	Same architecture, judges drawn only from families <i>not</i> in the evaluation suite; inter-judge agreement reported	JUDGED	low	Removes the judge/judged overlap directly, and answers 9W1B's request for intra-Tribunal agreement, which has never been reported.
E2	Off-the-shelf open-weight process reward model scoring each step	JUDGED	medium	Fully reproducible, no API, no frontier-family conflict. Open question — PRMs are trained on maths, not engineering, and whether that transfers would be a finding either way.
E3	Deterministic milestone verification from instrumented templates: extract (value, unit) pairs from the trace, check coverage of required intermediate quantities with unit normalization	SYMBOLIC	high	Takes models out of the critical path. Order-free, so indifferent to <i>how</i> a model reached a quantity.
E4	E3 plus symbolic checking — parse the model's stated equations, test algebraic equivalence and dimensional consistency against the gold formula with sympy	SYMBOLIC	high	Could catch right-number-wrong-reason, which pure value matching cannot.
E5	E3 as primary, non-family LLM adjudication only on residual unmatched steps	HYBRID	high	Deterministic where possible, judged where necessary — and the judged fraction becomes a reportable statistic. Possibly the most realistic option.

On instrumentation scope: only the frozen pilot templates would need refactoring to emit {id, symbol, value, unit, formula, depends_on, is_milestone}. Nothing else gets touched until there's evidence.

05

Experiments worth running

ID	EXPERIMENT	SUGGESTED DESIGN
X1	Meta-evaluation bake-off	Each evaluator's agreement with the human gold set — Spearman ρ and Kendall τ at trace level, step-level F1 and first-error-position accuracy at step level. Bootstrap CIs ($B = 10,000$). Pairwise significance via Williams' test for dependent correlations rather than two separate CIs. Results reported <i>against the human ceiling</i> rather than in absolute terms.

ID	EXPERIMENT	SUGGESTED DESIGN
X2	Self-preference / family overlap	Speaks directly to yAYU #1. Fixed trace set, 5 generator families × 4 judges. Estimate score(judge J, trace from family F); compare the J = F diagonal against the off-diagonal via the interaction term in a mixed model. Placebo: rerun with randomly permuted family labels. Blind condition: paraphrase traces to strip stylistic fingerprints — if the effect vanishes, it was cue-driven. Output would be a bias estimate <i>in Reasoning-F1 points with a CI</i> . A null result would settle the objection as well as a positive one.
X3	Leave-one-judge-out	Already promised in the July rebuttal. Recompute Tier 2 with each judge excluded; report $\Delta F1$ per evaluated model against the placebo and the untouched non-family models. Comes almost free once X2's infrastructure exists.
X4	Threshold sensitivity	FAC and Reasoning F1 across grids for the 2% numeric tolerance, the 0.7 cross-encoder threshold and the 0.8 alignment ratio. Promised; only τ has ever been varied.
X5	Variance and significance	Per-template SD across the 15 seeds as a Table 1 column; bootstrap CIs on headline claims; paired tests for the FAC-vs-F1 decoupling (McNemar for binary FAC, Wilcoxon signed-rank for continuous F1). Promised.

Would produce: the evidence any instrumentation decision rests on · every outstanding camera-ready promise discharged with real numbers.

Where the evidence would decide

SUGGESTED CRITERION

Do E3/E4/E5 beat E0 and E1 on agreement with the human gold set, with non-overlapping confidence intervals?

If yes — instrument more broadly
Refactor the remaining templates, re-score the model suite, and rewrite §4 around deterministic verification, so the title and the mechanism agree.

If no — ship the best judged variant
Take E1 or E2 with the bias analysis, threshold grids, ceiling and CIs. A weaker mechanism, but every statistical objection from both cycles answered with evidence — and a negative instrumentation result is itself reportable.

Both branches lead somewhere publishable, which seems worth preserving: no outcome where the effort produces nothing new to say about verification.

Objections, and what might answer each

Across both cycles. A row with nothing against it would ship unanswered — perhaps worth checking before submitting.

REVIEWER	OBJECTION	POSSIBLY ANSWERED BY
yAYU 1 · 9W1B 2 · gFWV 4	Circularity — judges share families with evaluated models; framework validated by the same kind of system it uses	X2 quantifies it, X3 tests removal, E1/E3/E5 remove the mechanism
yAYU 3 · 9W1B 2 · gFWV	$\rho = 0.632$ rests on $n = 100$, no CI, and is "only moderate" in absolute terms	Workstream 03 (larger n + ceiling), X1 (bootstrap CIs, Williams' test)
yAYU 2 · 9W1B 4	Single-run point scores for 27 models; no seeds, variance or significance behind headline claims	X5
yAYU 4 · 9W1B 3	Hard-coded thresholds (2% / 0.7 / 0.8) never varied	X4 , plus thresholds parameterized in the harness
9W1B 3	Tier-2 step scoring is coarse — 1.0 / 0.5 / 0.0 with a lowest-score tie-break	E3/E4 would replace it with continuous milestone coverage

REVIEWER	OBJECTION	POSSIBLY ANSWERED BY
9W1B (comment)	Inter-judge agreement within the Tribunal never reported, separately from human alignment	E1 reports it as standard output; X2 decomposes it by family
9W1B 6	Validation buried in appendices; central claims unassessable from the main text	Rewrite — ceiling, CIs and bias estimate moved into §4
gFWV (comment)	BERTScore can't discriminate reasoning quality in technical domains	Rewrite — demote textual metrics out of the headline table
cqGs 1 · ynoK 3	Tool-free setting conflates arithmetic execution with engineering reasoning; no tool/RAG baselines	<i>Probably out of scope this cycle.</i> E3/E4 separate arithmetic from structure by construction, which strengthens the argument without substituting for a baseline — likely stays in Limitations.

06 — RISKS

What would most likely go wrong

RISK	POSSIBLE MITIGATION
Expert annotation runs late. Probably the largest schedule risk — most other work waits on it.	Begin recruitment before anything else. Batch the labelling so partial returns remain usable, and label the triple-overlap subset <i>first</i> so the ceiling exists even if the remainder arrives late.
Open PRMs don't transfer from mathematical to physically-grounded reasoning.	That would be a reportable finding and a contribution to the process-supervision literature — worth budgeting for as an outcome rather than a failure.
Instrumentation cost per template exceeds estimate.	This is what the pilot slice is for: a reliable per-template cost estimate before committing to the rest.
Deterministic extraction proves brittle on messy model prose — missed values read as missed reasoning.	E5 hybrid is the designed mitigation. Extraction recall could also be measured against human step labels directly, so parsing failure stays separable from reasoning failure.
Milestone verification over-penalizes genuinely valid alternative derivations — the failure mode that would make the approach unusable.	This is why the gold standard would carry a "correct-but-alternative-path" label. Measure E3/E4 specifically on that subset; if they fail there, E5 or E1 wins on the evidence.

07 — WHERE TO BEGIN

Four suggested starting points

Begin expert recruitment

Likely the longest lead time and the one item whose duration you don't control, so probably worth starting before anything else.

Recover the generations

Pull the archived inference results for all 27 models and verify integrity against the 1,350-record expectation.

Freeze a pilot slice

Templates, problems and trace-source models chosen once and not renegotiated later, so every comparison rests on the same footing.

Read three papers

ProcessBench, Panickssery et al. on self-preference, and Deutsch et al. on metric meta-evaluation — they bear on most of the design choices above.

Suggestions drafted 5 September 2026 against the ARR May 2026 submission (docs/_ARR_May__EngTrace.pdf) and the reviewer sets from the January and July 2026 cycles. Offered for discussion; nothing here is settled.